

# APS110 Tutorial #3

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# Practice Question 1

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1. Determine which material below fits the following descriptions best
- a) The maximum length that can be reached by applying tension on a 50 cm metal bar before it begins to plastically deform is 50.04 cm
  - b) The material with the lowest strain at yield strength
  - c) The material with the highest strain at yield strength
  - d) The required applied force to plastically deform a tensile specimen with a rectangular cross-sectional area (10 mm x 5 mm) is 22.5 kN

| Material  | Yield Strength (MPa) | Modulus of Elasticity (GPa) |
|-----------|----------------------|-----------------------------|
| Tungsten  | 550                  | 407                         |
| Steel     | 180                  | 207                         |
| Nickel    | 138                  | 190                         |
| Titanium  | 450                  | 107                         |
| Brass     | 77                   | 97                          |
| Aluminum  | 35                   | 69                          |
| Magnesium | 130                  | 45                          |

# Practice Question 1

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$$\varepsilon = (l_i - l_0) / l_0 = (50.04 - 50) / 50 = 0.0008$$

Find  $\varepsilon$  for all and compare:

$$\varepsilon = \sigma / E \Rightarrow \text{a) } \varepsilon_{\text{Brass}} = (77\text{e}6) / (97\text{e}9) = 0.000793 \approx 0.0008 \quad \square \text{ Brass}$$

$$\text{b) } \varepsilon_{\text{Al}} = 0.000507 \text{ (lowest)} \quad \square \text{ Aluminum}$$

$$\text{c) } \varepsilon_{\text{Ti}} = 0.00420 \text{ (highest)} \quad \square \text{ Titanium}$$

$$\text{d) } \sigma = F / A = 22.5\text{e}3 / (10\text{e-}3 \cdot 5\text{e-}3) = 450 \text{ MPa} \quad \square \text{ Titanium}$$

# Practice Question 2

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2. A bar of steel is loaded by 960 N in three-point bending. The bar has a length of 70 cm, a width of 4 cm, and a height of 1.5 cm. This sample has a yield strength of 515 MPa.
- a) Will this bar plastically deform?
  - b) At what force would this bar plastically deform?

a)  $\sigma = \frac{3FL}{2bd^2} = \frac{3 \cdot 960 \cdot 70e-2}{2 \cdot 4e-2 \cdot (1.5e-2)^2} = 112 \text{ MPa} < 515 \text{ MPa}, \therefore \text{Will not plastically deform}$

b)  $F = \frac{2\sigma bd^2}{3L} = \frac{2 \cdot 515e6 \cdot 4e-2 \cdot (1.5e-2)^2}{3 \cdot 70e-2} = 4.41 \text{ kN}$

# Practice Quiz

ceiling must be 30 cm, and the total supported weight is 1000N. To keep the costs as low as possible, you will be using a 1020 steel alloy with a modulus of elasticity of 207 GPa and a yield strength of 295 MPa.

- a) What is the minimum diameter that you can design this tie such that it will not undergo plastic deformation?
  - b) If another tie was designed with a diameter of 2.3 cm, how long would this tie be just prior to permanently deforming?
2. A load  $F$  (in N) is to be suspended from a cylindrical length of line of radius  $R$  (in m) and unloaded length  $l_0$  (in m) and modulus  $E$  (in Pa) without breaking or plastically deforming the line. Develop an equation in terms of these variables for the change in length that the line will undergo when a weight is added to the end.
3. A hypothetical ceramic was tested utilizing three-point bending. The sample has a cross-section with a height of 7 mm and a width of 10 mm. This sample broke when a load of 5000 N was applied to it with a distance between the support points of 50 mm.
- a) Another test is to be performed on a sample made of the same ceramic material with a cross-section with a height of 5 mm and a width of 7.5 mm. If the span between the support points is maintained at 50 mm, at what load would this second sample fracture?
  - b) Discuss reasons why ceramics are tested using 3-point bending as opposed to a tensile test.
4. A bar of steel with a Young's modulus of 210 GPa was loaded onto a tensile test machine. After a load is applied to its long axis, the bar was elongated by 100 mm. The initial cross section of the bar is rectangular with a width of 0.005 m and a height of 0.02 m. The initial length of the sample is 1 m. What is the applied load?
- a) 2.1 kN
  - b) 21 kN
  - c) 210 kN

$$1. \quad \left. \begin{array}{l} \text{a)} \\ \text{b)} \end{array} \right\} \sigma = \frac{F}{A} = \frac{F}{\pi \left(\frac{D}{2}\right)^2} = \frac{4F}{\pi D^2} \Rightarrow D = \left[ \frac{4F}{\pi \sigma} \right]^{\frac{1}{2}} = \left[ \frac{4 \cdot 1000}{\pi \cdot 295e6} \right]^{\frac{1}{2}} = 2.1 \text{ mm}$$

$$\text{b)} \quad \varepsilon = \frac{l_i - l_0}{l_0} \Rightarrow l_i = \varepsilon l_0 + l_0$$

$$2. \quad \varepsilon = \frac{\sigma}{E} = \frac{295e6}{207e9} = 1.425e^{-3} \rightarrow l_i = (1.425e^{-3} \cdot 30) + 30 = 30.04 \text{ cm}$$

$$2. \quad \left. \begin{array}{l} \sigma = E\varepsilon \Rightarrow \varepsilon = \sigma/E \\ \varepsilon = \Delta l / l_0 \Rightarrow \Delta l = \varepsilon l_0 \\ \sigma = F/A \text{ where } A = \pi R^2 \\ \sigma = E\varepsilon \Rightarrow \varepsilon = \sigma/E \end{array} \right\} \rightarrow \begin{aligned} \Delta l &= \varepsilon l_0 \\ &= \frac{\sigma l_0}{E} \\ &= \frac{F l_0}{AE} \\ &= \frac{F l_0}{E \pi R^2} \end{aligned}$$

3. a) Sample 1:  $\sigma_f = \frac{3FL}{2wh^2} = \frac{3 \cdot 5000 \cdot 50e-3}{2 \cdot 10e-3 \cdot (7e-3)^2} = 765 \text{ MPa}$

Sample 2:  $F = \frac{2\sigma_f wh^2}{3L} = \frac{2 \cdot 765e6 \cdot 7.5e-3 \cdot (5e-3)^2}{3 \cdot 50e-3} = 1.91 \text{ kN}$

- b) Top three reasons: 1. Can crumple in grip  
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 2. Difficult to machine/form into complex specimen shape  
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 3. Difficult to align in standard tensile test which rely on alignment  
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4.  $\sigma = F/A, \sigma = E\varepsilon \Rightarrow F = E\varepsilon A$

$\varepsilon = \Delta l / l_0 \Rightarrow F = (E \cdot A \cdot \Delta l) / l_0 = (100e-3 \cdot 210e9 \cdot 0.005 \cdot 0.02) / 1 = 2100 \text{ kN}$